Database Technology - Assignment 5

Note: To solve below queries, use “hr” database

1. Display all the first names in lower case and the last names in upper case as the single column separated with '
- '. Provide the meaningful name to the column.

```
mysql> SELECT CONCAT(LOWER(first_name), ' - ', UPPER(last_name)) AS full_name
    -> FROM employees;
+---------------------+
| full_name           |
+---------------------+
| ellen - ABEL        |
| sundar - ANDE       |
| mozhe - ATKINSON    |
| shelli - BAIDA      |
| amit - BANDA        |
```

```
| clara - VISHNEY     |
| shanta - VOLLMAN    |
| alana - WALSH       |
| matthew - WEISS     |
| jennifer - WHALEN   |
| david - WILLIAMS    |
| neena - YANG        |
| eleni - ZLOTKEY     |
+---------------------+
107 rows in set (0.001 sec)
```

2. Display the first word of the job title. (Hint : Use substring_index).

```
mysql> SELECT SUBSTRING_INDEX(JOB_TITLE,' ', 1) FROM jobs;
+-----------------------------------+
| SUBSTRING_INDEX(JOB_TITLE,' ', 1) |
+-----------------------------------+
| Public                            |
| Accounting                        |
| Administration                    |
| President                         |
| Administration                    |
| Accountant                        |
| Finance                           |
| Human                             |
| Programmer                        |
| Marketing                         |
| Marketing                         |
| Public                            |
| Purchasing                        |
| Purchasing                        |
| Sales                             |
| Sales                             |
| Shipping                          |
| Stock                             |
| Stock                             |
+-----------------------------------+
19 rows in set (0.001 sec)
```

3. Display the First name, last name and length of first name for the employees whose last name contains character ‘b’ anywhere after 3rd position.

```
mysql> SELECT FIRST_NAME, LAST_NAME, LENGTH(FIRST_NAME) FROM employees WHERE SUBSTR(LAST_NAME, 4)
 LIKE '%b%';
+------------+-----------+--------------------+
| FIRST_NAME | LAST_NAME | LENGTH(FIRST_NAME) |
+------------+-----------+--------------------+
| Gerald     | Cambrault |                  6 |
| Nanette    | Cambrault |                  7 |
| Nancy      | Gruenberg |                  5 |
| Susan      | Jacobs    |                  5 |
+------------+-----------+--------------------+
4 rows in set (0.002 sec)
```

4. Display the employee id, first name and last name of the employees whose first or last name includes a space in between.

```
107 rows in set (0.001 sec)

mysql> SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME FROM EMPLOYEES WHERE FIRST_NAME LIKE '% %' OR LAST_NAME LIKE '% %';
+-------------+-------------+-----------+
| EMPLOYEE_ID | FIRST_NAME  | LAST_NAME |
+-------------+-------------+-----------+
|         112 | Jose Manuel | Urman     |
+-------------+-------------+-----------+
1 row in set (0.001 sec)

mysql> SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME FROM EMPLOYEES WHERE LOCATE(' ', FIRST_NAME)>0 OR LOCATE(' ', LAST_NAME)>0;
+-------------+-------------+-----------+
| EMPLOYEE_ID | FIRST_NAME  | LAST_NAME |
+-------------+-------------+-----------+
|         112 | Jose Manuel | Urman     |
+-------------+-------------+-----------+
1 row in set (0.001 sec)
```

5. Display first name, salary, and round the salary to thousands.

```
mysql> SELECT FIRST_NAME, salary, ROUND(salary, -3) FROM employees;
+-------------+----------+-------------------+
| FIRST_NAME  | salary   | ROUND(salary, -3) |
+-------------+----------+-------------------+
| Steven      | 24000.00 |             24000 |
| Neena       | 17000.00 |             17000 |
| Lex         | 17000.00 |             17000 |
| Alexander   |  9000.00 |              9000 |
| Bruce       |  6000.00 |              6000 |
| David       |  4800.00 |              5000 |
| Valli       |  4800.00 |              5000 |
```

```
| Donald      |  2600.00 |              3000 |
| Douglas     |  2600.00 |              3000 |
| Jennifer    |  4400.00 |              4000 |
| Michael     | 13000.00 |             13000 |
| Pat         |  6000.00 |              6000 |
| Susan       |  6500.00 |              7000 |
| Hermann     | 10000.00 |             10000 |
| Shelley     | 12008.00 |             12000 |
| William     |  8300.00 |              8000 |
+-------------+----------+-------------------+
107 rows in set (0.001 sec)
```

6. Display employee ID and the date on which he ended his previous job in the format as mentioned. (Format: Thursday, October 27th 2011)

```
mysql> SELECT EMPLOYEE_ID, END_DATE, DATE_FORMAT(END_DATE, '%W, %M %D %Y') END_DATE_FORMATTED FRO
M JOB_HISTORY;
+-------------+------------+----------------------------+
| EMPLOYEE_ID | END_DATE   | END_DATE_FORMATTED         |
+-------------+------------+----------------------------+
|         101 | 2011-10-27 | Thursday, October 27th 2011 |
|         101 | 2015-03-15 | Sunday, March 15th 2015     |
|         102 | 2016-07-24 | Sunday, July 24th 2016      |
|         114 | 2017-12-31 | Sunday, December 31st 2017  |
|         122 | 2017-12-31 | Sunday, December 31st 2017  |
|         176 | 2016-12-31 | Saturday, December 31st 2016 |
|         176 | 2017-12-31 | Sunday, December 31st 2017  |
|         200 | 2011-06-17 | Friday, June 17th 2011      |
|         200 | 2016-12-31 | Saturday, December 31st 2016 |
|         201 | 2017-12-19 | Tuesday, December 19th 2017 |
+-------------+------------+----------------------------+
10 rows in set (0.001 sec)
```

7. Display first name and date of first salary of the employees. (Consider 1st day of month as salary day. HINT: LAST_DAY).

```
mysql> SELECT FIRST_NAME, HIRE_DATE, DATE_ADD(LAST_DAY(HIRE_DATE), INTERVAL 1 DAY) AS FIR
ST_SAL FROM EMPLOYEES;
+-------------+------------+------------+
| FIRST_NAME  | HIRE_DATE  | FIRST_SAL  |
+-------------+------------+------------+
| Steven      | 2013-06-17 | 2013-07-01 |
| Neena       | 2015-09-21 | 2015-10-01 |
| Lex         | 2011-01-13 | 2011-02-01 |
| Alexander   | 2016-01-03 | 2016-02-01 |
| Bruce       | 2017-05-21 | 2017-06-01 |
| David       | 2015-06-25 | 2015-07-01 |
| Valli       | 2016-02-05 | 2016-03-01 |
| Diana       | 2017-02-07 | 2017-03-01 |
| Nancy       | 2012-08-17 | 2012-09-01 |
| Daniel      | 2012-08-16 | 2012-09-01 |
```

```
| Britney     | 2015-03-03 | 2015-04-01 |
| Samuel      | 2016-07-01 | 2016-08-01 |
| Vance       | 2017-03-17 | 2017-04-01 |
| Alana       | 2016-04-24 | 2016-05-01 |
| Kevin       | 2016-05-23 | 2016-06-01 |
| Donald      | 2017-06-21 | 2017-07-01 |
| Douglas     | 2018-01-13 | 2018-02-01 |
| Jennifer    | 2013-09-17 | 2013-10-01 |
| Michael     | 2014-02-17 | 2014-03-01 |
| Pat         | 2015-08-17 | 2015-09-01 |
| Susan       | 2012-06-07 | 2012-07-01 |
| Hermann     | 2012-06-07 | 2012-07-01 |
| Shelley     | 2012-06-07 | 2012-07-01 |
| William     | 2012-06-07 | 2012-07-01 |
+-------------+------------+------------+
107 rows in set (0.001 sec)
```

8. Display first name and experience of the employees in the format of number of years, months and days completed.

```
mysql> SELECT first_name, TIMESTAMPDIFF(YEAR, HIRE_DATE, NOW()) YRS, TIMESTAMPDIFF(MONTH,HIRE_DAT
E,NOW())%12 MTHS, DATEDIFF( NOW(), ADDDATE(ADDDATE(HIRE_DATE, INTERVAL TIMESTAMPDIFF(YEAR,HIRE_DA
TE,NOW())YEAR),INTERVAL TIMESTAMPDIFF(MONTH, HIRE_DATE, NOW())%12 MONTH))DAYS FROM EMPLOYEES;
+-------------+------+------+------+
| first_name  | YRS  | MTHS | DAYS |
+-------------+------+------+------+
| Steven      |   12 |    8 |   12 |
| Neena       |   10 |    5 |    8 |
| Lex         |   15 |    1 |   16 |
| Alexander   |   10 |    1 |   26 |
| Bruce       |    8 |    9 |    8 |
| David       |   10 |    8 |    4 |
| Valli       |   10 |    0 |   24 |
| Diana       |    9 |    0 |   22 |
| Nancy       |   13 |    6 |   12 |
| Daniel      |   13 |    6 |   13 |
| John        |   10 |    5 |    1 |
```

```
| Samuel      |    9 |    8 |    0 |
| Vance       |    8 |   11 |   12 |
| Alana       |    9 |   10 |    5 |
| Kevin       |    9 |    9 |    6 |
| Donald      |    8 |    8 |    8 |
| Douglas     |    8 |    1 |   16 |
| Jennifer    |   12 |    5 |   12 |
| Michael     |   12 |    0 |   12 |
| Pat         |   10 |    6 |   12 |
| Susan       |   13 |    8 |   22 |
| Hermann     |   13 |    8 |   22 |
| Shelley     |   13 |    8 |   22 |
| William     |   13 |    8 |   22 |
+-------------+------+------+------+
107 rows in set (0.003 sec)

mysql>
```

9. Display the employee id, first name and joining date of the employees who joined in the month of august of any year.

```
mysql> SELECT employee_id, first_name, HIRE_DATE FROM EMPLOYEES WHERE MONTH(HIRE_DATE) = 08;
+-------------+------------+------------+
| employee_id | first_name | HIRE_DATE  |
+-------------+------------+------------+
|         108 | Nancy      | 2012-08-17 |
|         109 | Daniel     | 2012-08-16 |
|         119 | Karen      | 2017-08-10 |
|         129 | Laura      | 2015-08-20 |
|         134 | Michael    | 2016-08-26 |
|         152 | Peter      | 2015-08-20 |
|         158 | Allan      | 2014-08-01 |
|         189 | Jennifer   | 2015-08-13 |
|         202 | Pat        | 2015-08-17 |
+-------------+------------+------------+
9 rows in set (0.001 sec)

mysql>
```

10. Display the details of employees who joined before the year 2015.

```
mysql> SELECT EMPLOYEE_ID, FIRST_NAME, LAST_NAME, HIRE_DATE FROM EMPLOYEES WHERE YEAR(HIRE_DATE) <
 2015;
+-------------+------------+-----------+------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | HIRE_DATE  |
+-------------+------------+-----------+------------+
|         100 | Steven     | King      | 2013-06-17 |
|         102 | Lex        | Garcia    | 2011-01-13 |
|         108 | Nancy      | Gruenberg | 2012-08-17 |
|         109 | Daniel     | Faviet    | 2012-08-16 |
|         114 | Den        | Li        | 2012-12-07 |
|         115 | Alexander  | Khoo      | 2013-05-18 |
|         120 | Matthew    | Weiss     | 2014-07-18 |
|         122 | Payam      | Kaufling  | 2013-05-01 |
|         133 | Jason      | Mallin    | 2014-06-14 |
|         137 | Renske     | Ladwig    | 2013-07-14 |
|         141 | Trenna     | Rajs      | 2013-10-17 |
|         145 | John       | Singh     | 2014-10-01 |
|         156 | Janette    | King      | 2014-01-30 |
|         157 | Patrick    | Sully     | 2014-03-04 |
|         158 | Allan      | McEwen    | 2014-08-01 |
|         174 | Ellen      | Abel      | 2014-05-11 |
|         184 | Nandita    | Sarchand  | 2014-01-27 |
|         192 | Sarah      | Bell      | 2014-02-04 |
|         200 | Jennifer   | Whalen    | 2013-09-17 |
|         201 | Michael    | Martinez  | 2014-02-17 |
|         203 | Susan      | Jacobs    | 2012-06-07 |
|         204 | Hermann    | Brown     | 2012-06-07 |
|         205 | Shelley    | Higgins   | 2012-06-07 |
|         206 | William    | Gietz     | 2012-06-07 |
+-------------+------------+-----------+------------+
24 rows in set (0.001 sec)
```

11. Display the number of days between system date and 1st January 2011.

```
mysql> SELECT DATEDIFF(CURDATE(), '2011-01-01') NoOfDays;
+----------+
| NoOfDays |
+----------+
|     5538 |
+----------+
1 row in set (0.001 sec)
```

12. Display number of employees joined after 15th of any month.

```
mysql> SELECT COUNT(*) AS NoOfEmpl FROM EMPLOYEES WHERE DAY(HIRE_DATE) > 15;
+----------+
| NoOfEmpl |
+----------+
|       57 |
+----------+
1 row in set (0.003 sec)
```

13. Display third highest salary of employees.

```
mysql> SELECT DISTINCT SALARY FROM EMPLOYEES ORDER BY SALARY DESC LIMIT 2,1;
+----------+
| SALARY   |
+----------+
| 14000.00 |
+----------+
1 row in set (0.001 sec)
```